Fizika

9 sinf

MAVZU: Temperatura.

Issiqlik muvozanati

Temperatura – moddaning issiqlik darajasini miqdor jihatidan aniqlaydigan fizik kattalikdir. Temperatura molekulyar fizika va termodinamikaning asosiy kattaliklaridan biridir.

Temperatura lotinchada *“holat”* degan ma’noni anglatadi.

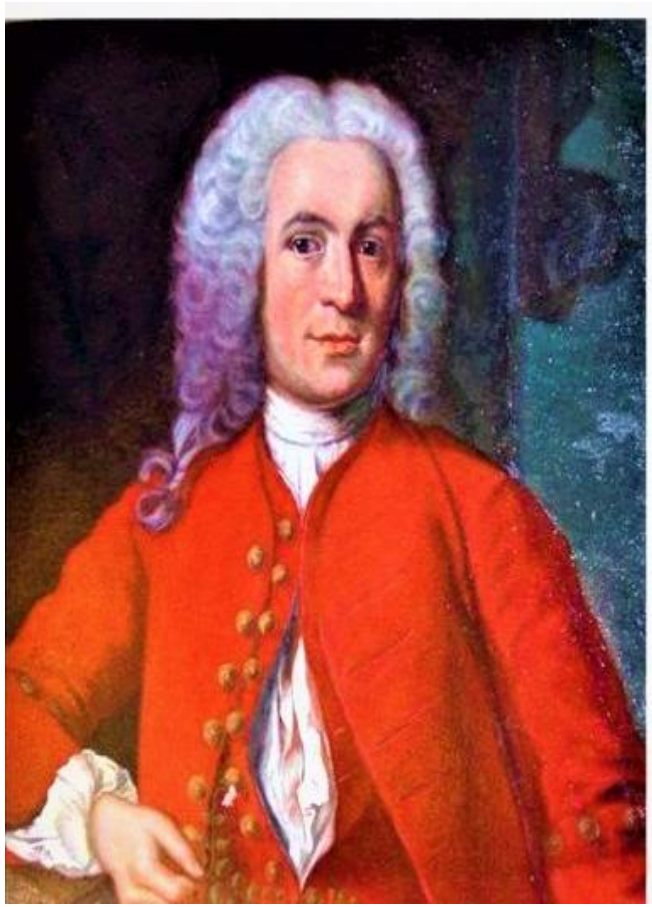
Moddalarda issiqlik almashinishi natijasida ularning temperaturalari tenglashishiga ***issiqlik muvozanati*** deyiladi.

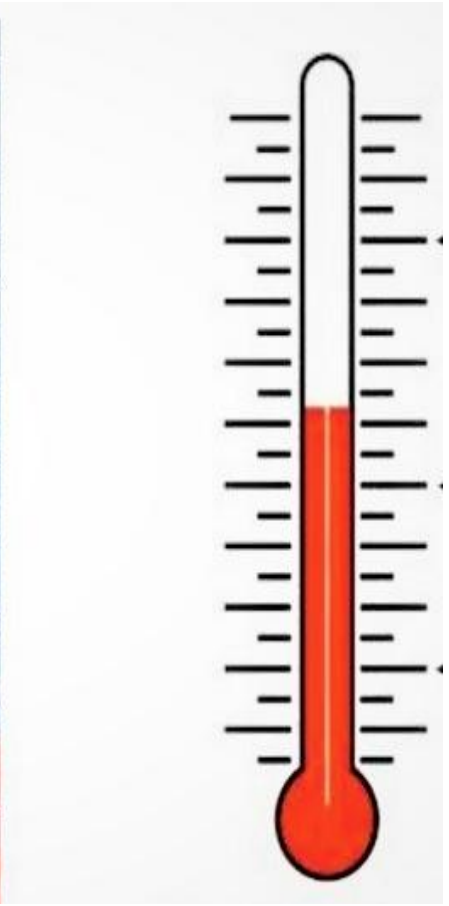
Temperaturaning Selsiy shkalasi

Temperatura termometr yordamida o‘lchanadi. Ko‘p foydalaniladigan termometr simobli termometrlardir. Termometr shkalasi darajalangan bo‘lib, simobning qancha ko‘tarilganligiga qarab temperaturani bilib olish mumkin. Temperaturaning o‘lchov birligi qilib ***gradus*** olingan. Gradus lotinchada “qadam” degan ma’noni anglatadi.

1ºC= 1 gradus Selsiy

Temperaturaning Selsiy shkalasi





1742-yilda shved olimi Andres Selsiy tomonidan ***Selsiy shkalasi*** yaratildi.

100ºC – suvning qaynash temperaturasi.

0ºC-muzning erish temperaturasi.

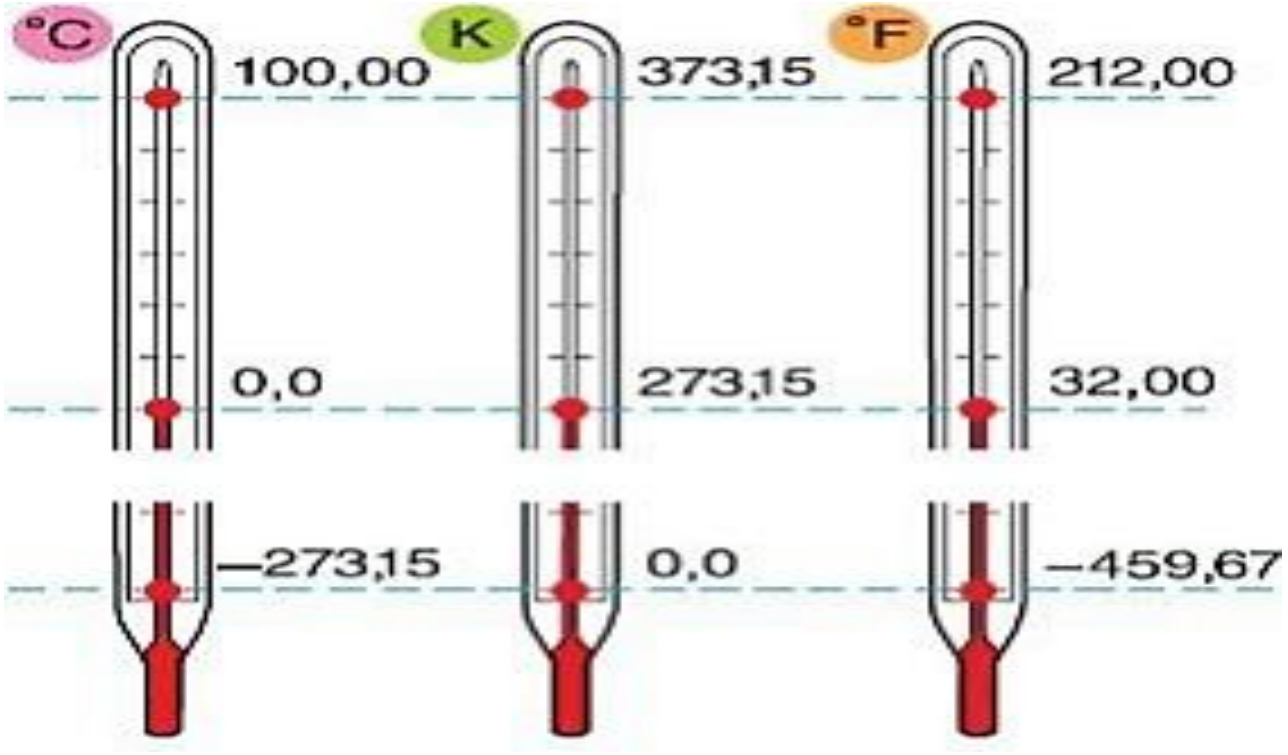
Absolyut temperatura

Ingliz olimi Uilyam Tompson (**Kelvin**) 1848-yilda temperaturaning absolyut shkalasini taklif qildi. Absolyut temperaturaning bu shkalasi **Kelvin shkalasi** deb ataldi.

Absolyut temperaturaning XBS dagi birligi *Kelvin* deb ataladi va “K” harfi bilan belgilanadi. Kelvin shkalasida olingan temperatura birligining qadamlari Selsiy shkalasining qadamlariga teng qilib olingan. Selsiy shkalasida o‘lchangan absolyut nol temperatura -273,15℃ ekanligi aniqlangan va buni taxminan -273℃ deb quyidagi formula keltirib chiqarilgan:

$$T = t + 273$$

Temperatura shkalalari



Selsiy shkalasi

Kelvin shkalasi

Farengeyt shkalasi

Temperaturaning molekulyar-kinetik talqini

Temperatura – gaz molekulalari ilgarilanma harakati o‘rtacha kinetik energiyasining o‘lchovidir. Molekulalarning tartibsiz harakati issiqlik harakati deb ataladi.

$$E_k = \frac{3}{2}\mathrm{kT}, \quad k = 1{,}38 \cdot 10^{-23}\ \mathrm{J/K}$$ *Bolsman doimiysi*.

Bolsman doimiysi – energiya birligi bilan temperatura birligi orasidagi munosabatni ifodalovchi kattalik.

Issiqlik muvozanati holatida barcha gaz molekulalarining ilgarilanma harakat o‘rtacha kinetik energiyasi (temperatura) bir xil bo‘ladi.

$p = \frac{2}{3} \cdot n \cdot \overline{E_k}$, $E_k = \frac{3}{2} kT$ dan

$p = \frac{2}{3} \cdot n \cdot \frac{3}{2} \cdot k \cdot T = n \cdot k \cdot T$ ya’ni $p = nkT$

Demak, ideal gazning bosimi gaz molekulalarining konsentratsiyasi va gaz temperaturasiga to‘g‘ri proporsional ekan.

5-mashq 1-masala

Quyidagi Selsiy shkalasida ifodalangan temperaturalarni Kelvin shkalasiga o‘tkazing:

0℃, 27℃, 100℃, 127℃, -73℃, -223℃, -200℃

Yechish:

$$T = t + 273$$

$$T_2 = 27℃ + 273 = 300\ \text{K}$$

$$T_6 = -223℃ + 273 = 50\ \text{K}$$

5-mashq 2-masala

Quyidagi Kelvin shkalalarida ifodalangan temperaturalarni Selsiy shkalasiga o‘tkazing:

0 K, 73 K, 273 K, 300 K, 773 K, 1000 K, 2000 K.

Yechish:

$$T = t + 273 \Rightarrow t = T - 273$$

$t_2 = 73\ \text{K} - 273 = -200℃$

$t_5 = 773\ \text{K} - 273 = 500℃$

5-mashq 5-masala

Idish ichidagi gaz temperaturasi 400 K bo‘lganda, manometr idishdagi bosim 276 kPa ga teng bo‘lganini ko‘rsatdi. Idishdagi gaz molekulalarining konsentratsiyasi nimaga teng?

Berilgan:

$T = 400\ \text{K} = 4 \cdot 10^2\ \text{K}$

$p = 276\ \text{kPa} = 2{,}76 \cdot 10^5 \text{Pa}$

$k = 1{,}38 \cdot 10^{-23}\ \text{J/K}$

Formula:

$p = n \cdot k \cdot T$

$n = \frac{p}{k \cdot T}$

Topish kerak: $n - ?$

Yechish: $n = \frac{2{,}76 \cdot 10^5 Pa}{1{,}38 \cdot 10^{-23} \frac{J}{K} \cdot 4 \cdot 10^2\ K} = 5 \cdot 10^{25} m^{-3}$

Javob: $\boldsymbol{n = 5 \cdot 10^{25} m^{-3}}$

Mustaqil bajarish uchun topshiriqlar

1. Temperaturaning qanday o‘lchov birliklarini bilasiz?

2. Selsiy va Kelvin temperaturasi o‘zaro qanday bog‘langan?

3. Gazning bosimi qanday fizik kattaliklarga bog‘liq?

4. 5 – mashq: 1-,2-masalaning davomini va 3-masalani yechish (24-bet).